

SQL Learning Journal #19 – SUBQUERYING

We can use subqueries to connect and filter different information from tables.

Example Tables

information_table

name	surname	hometown
John	Tone	Boston
Gunar	Blair	New_Jersey
Alex	Donna	New_York
Jonnes	Knutt	Califormia

students_table

name	surname	birth_date
John	Tone	1982
Gunar	Blair	1989
Alex	Donna	1975
Jonnes	Knutt	1984

1. SEMI JOIN (IN)

The code below can be used to filter information from students_table.

```
SELECT birth_date
FROM students_table
WHERE birth_date > 1980;
```

We can use the query above with another query to filter and select information from information_table.

```
SELECT name, surname, hometown
FROM information_table
WHERE birth_date IN
(
    SELECT birth_date
    FROM students_table
    WHERE birth_date > 1980
);
```

Result table

name	surname	hometown
John	Tone	Boston
Gunar	Blair	New_Jersey
Jonnes	Knutt	Califormia

2. ANTI JOIN (NOT IN)

This function works similarly to a semi join. However, an anti join selects information that is not included in the second table.

```
SELECT name, surname, hometown
FROM information_table
WHERE birth_date NOT IN
(
    SELECT birth_date
    FROM students_table
    WHERE birth_date > 1980
);
```

Result table

name	surname	hometown
Alex	Donna	New_York

What I Learned Today

Today I learned how subqueries can be used for more detailed filtering and for selecting information from different tables.